

Context-Aware Speed and Separation Monitoring for Human–Robot Ceiling-Panel Installation: An Exploratory Evaluation

Zenan Wu, Luke Siniakov, and Michael Magila

Abstract—Overhead panel installation requires a worker and a robot to coordinate around a shared payload, a setting in which fixed protective zones cannot distinguish approaching, working and retreating behaviour and therefore interrupt the task more often than necessary. This paper presents a context-aware speed and separation monitoring (SSM) controller for human–robot ceiling-panel installation and its evaluation on a physical prototype. A three-state Gaussian-mixture hidden Markov model (HMM) recognises the task phase from motion-capture features, a kinematic predictor detects rapid intrusion toward the robot, and a deterministic command ceiling guarantees that neither learned output can raise the robot speed above the reactive separation envelope. In leave-one-operator-out validation the causal phase recogniser reached 0.794 accuracy and 0.725 balanced accuracy. Twelve participants completed a counterbalanced three-block protocol with fixed, reactive and predictive controllers; 14 telemetry trials and 31 block questionnaires were analysed. Participants reported high confidence that the robot would stop and low workload, and mean perceived safety rose from 3.82/5 before the session to 4.45/5 after it. The telemetry showed that the choice of geometric reference and the execution of cued movements materially change how the recorded control behaviour should be read. The study demonstrates the integration of task-phase recognition, intrusion prediction and multimodal sensing in a construction-style task, and identifies the measurement requirements for a conclusive comparison of the three controllers.

Index Terms—Human–robot collaboration, speed and separation monitoring, hidden Markov model, construction robotics, motion capture, human factors.

I. INTRODUCTION AND BACKGROUND

OVERHEAD panel installation combines handling, alignment and fastening in a workspace shared by a worker and a moving payload. A robot can support the panel while the worker supplies judgement and dexterity. However, the worker’s position changes during the task, the payload enlarges the occupied volume, and unnecessary stops can disrupt coordination.

Speed and separation monitoring (SSM) depends on protective separation and timely protective action [1]–[3]. A fixed threshold is transparent but does not distinguish approaching, stationary and retreating motion. Task context may improve coordination, provided recognition errors cannot relax the deterministic command bound. The controller presented here therefore separates task-phase recognition, kinematic prediction and geometric constraints.

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The research question is whether task context can improve coordination beyond fixed-zone and reactive policies while remaining bounded by the same geometric constraint. The intended comparison considers intrusion response, unnecessary interruption, task efficiency and participant experience. The present exploratory analysis addresses three supporting questions: how accurately the model recognises task phase under causal operation; how geometry and cued movement affect the interpretation of recorded control behaviour; and what the questionnaire responses show about the experience of successive task blocks. These questions characterise the prototype and the evidence needed to evaluate the original comparative aim.

The work contributes a physical ceiling-panel collaboration prototype that separates task-phase recognition from kinematic intrusion prediction under an explicit command ceiling; a descriptive analysis of geometric inputs and robot motion during the task; and item-level human-experience results with a complete-response sensitivity analysis. The contribution is the application-specific integration and evaluation of these elements rather than a new SSM formulation.

II. RELATED WORK

A. Construction HRC

Robotic construction research has increasingly moved from isolated automation toward systems in which human expertise and robot strength are combined [4], [5]. Reviews of construction HRC identify safety, perception, communication and task allocation as continuing barriers to deployment [6], [7]. Panel installation is an appropriate case study because the person and robot must coordinate around a shared object and because excessive conservative stopping directly affects task fluency.

B. Speed and Separation Monitoring

SSM requires more than a fixed radius around the robot. The required separation depends on human and robot motion during sensing, controller reaction and robot stopping, together with intrusion and position-uncertainty allowances [2], [3]. Byner et al. [8] derive robot velocity limits using separation and motion direction, and evaluate cycle performance in a physical machine-tending task. Karagiannis et al. [9] switch safety-zone configurations according to robot position, using an overhead safety camera, a safety-rated programmable logic

controller (PLC) and the robot controller in an automotive-inspired cell. These studies establish that adaptation and productivity evaluation are already part of SSM research, and they define the reactive comparator against which task context must show incremental value.

Construction-specific work further narrows the contribution. Xiao Lin et al. [10] integrate worker, robot and material geometry in a digital twin with dynamic protective-distance calculation, evaluating static and dynamic conflict scenarios. Their treatment of manipulated materials is particularly relevant to panel installation, because a proxy around the robot alone does not describe the occupied volume once a panel is carried. Table I summarises how these approaches relate to the present prototype.

C. Human Activity Recognition and Collaboration

Hidden Markov models (HMMs) provide an interpretable probabilistic representation of temporal state. They describe a hidden state sequence through a transition matrix and relate each state to observable measurements through an emission model [11]. HMMs have been used in wearable activity recognition and human-robot task modelling [12], [13]. However, task phase and hazard are different variables: a person can trip while approaching, working or retreating. This project therefore uses the HMM only for task context. The rapid-intrusion event is detected separately by the kinematic intrusion predictor of Section III-D, which uses only measured distance and closing speed.

Semantic states such as “working” or “retreating” describe task activity, whereas a hazardous situation concerns exposure to harm. A rapid approach may occur within any phase, so treating hazard as a mutually exclusive phase would confound activity with event recognition. Separating these variables keeps the geometric command bound testable independently of the HMM’s interpretation.

A closely related construction study by Xiao Lin et al. [14] combines a semantic workflow model, policies learned from demonstration and local safety control in a virtual-reality wall-panel task with simulated sensors. The present study complements that work by evaluating a physical ceiling-panel prototype with measured robot telemetry and participant questionnaires.

Objective collaboration fluency is commonly evaluated through temporal measures such as human idle time, robot idle time, concurrent activity and functional delay [15]. Perceived workload and trust can add explanatory evidence, but subjective reports cannot establish physical safety or replace measured response [16], [17].

The evaluation in this paper therefore connects task recognition, measured control inputs and perceived experience in a physical panel task.

III. SYSTEM DESIGN

A. Overview

The controller (Fig. 1) takes as input the position and velocity of the worker relative to the robot, obtained from motion capture, and the robot’s tool position, obtained from

the robot controller. Its output is a single speed fraction in $[0, 1]$ that is applied to the robot’s speed override, with a pause request when the fraction is zero. The remainder of this section describes the apparatus, the geometric features, the task-context and intrusion models, and the policies that map these to the command.

B. Apparatus and Sensing

The laboratory prototype (Fig. 2) uses a Universal Robots UR10 six-axis arm fitted with a vacuum gripper that carries a lightweight ceiling-panel surrogate. The arm lifts the panel to ceiling height, the participant approaches to work on the raised panel, and the arm lowers it again. The participant’s motion is measured by two independent systems.

Xsens MVN is a wearable inertial motion-capture suit. Seventeen small inertial measurement units, each containing an accelerometer, gyroscope and magnetometer, are strapped to body segments. Their signals are fused with a biomechanical model into the position and orientation of 23 body segments (pelvis, spine, head, arms, legs and feet) at up to 240 Hz [18], [19]. Because position is obtained by integrating inertial signals, the estimate drifts slowly in the world frame, and its origin is set by a calibration pose rather than by the room. In this work the Xsens path supplies the body-segment origins used as the controller’s body-distance input in the later geometry version.

OptiTrack is an optical motion-capture system. Several infrared cameras mounted around the cell (visible on the frame in Fig. 2a) observe passive reflective markers, and the Motive software triangulates each marker to millimetre-level position in a fixed room coordinate frame [20]. A cluster of markers on the participant’s head cap is tracked as a single rigid body. OptiTrack therefore supplies an absolute head position relative to the robot without drift, but it does not describe the rest of the body. In this work the OptiTrack head position is the input to the distance and closing-speed features that drive the phase model and the reactive envelope.

The two systems are combined by role rather than by fusing their estimates: OptiTrack provides the absolute head position relative to the robot, and Xsens provides body posture. Both streams enter the same feature extractor through a common sample interface, so the controller is independent of which tracker supplied a sample. Robot telemetry from the UR controller supplies the tool centre point (TCP) pose, that is the position of the gripper face, and the six joint velocities. The control loop runs at a nominal 60 Hz and the actual sample intervals are logged.

C. Geometry Versions

For a tracked point $\mathbf{p}$ (the head position) and the nearest point $\mathbf{q}$ on a vertical column proxy that stands in for the robot, the head features are

$$d = \|\mathbf{q} - \mathbf{p}\|_2, \quad v_{\text{proj}} = \mathbf{v}^T \mathbf{u}, \quad (1)$$

where d is the separation distance in metres, $\mathbf{v}$ is the head velocity estimated from the slope of recent samples, $\mathbf{u}$ is the unit vector from the head toward the proxy, and v_{proj} is

TABLE I
SELECTED RELATED METHODS AND THEIR RELATIONSHIP TO THE PRESENT STUDY.

Study	Method and evaluation setting	Implication for this study
Byner et al. [8]	Continuous velocity limits from separation and motion direction; physical machine-tending cycle evaluation.	Reactive speed adaptation is an established comparator; phase recognition must demonstrate incremental value.
Karagiannis et al. [9]	Robot-position-dependent zone switching; overhead safety camera and safety PLC in an automotive-inspired cell.	A software command bound is distinct from a safety architecture implemented with safety components.
X. Lin et al. [10]	Dynamic protective distance with worker, robot and material geometry; construction conflict scenarios.	Panel and body geometry require validation beyond point-proxy arithmetic consistency.
X. Lin et al. [14]	Learned semantic workflow and local safety control; VR wall-panel installation with simulated sensors.	Task-aware safety control already exists; physical telemetry and reported experience define the present evaluation focus.
Present study	Causal HMM phase context and kinematic prediction beneath a command ceiling; physical ceiling-panel prototype.	Telemetry and questionnaire evaluation of a construction-style task with a human participant in the loop.

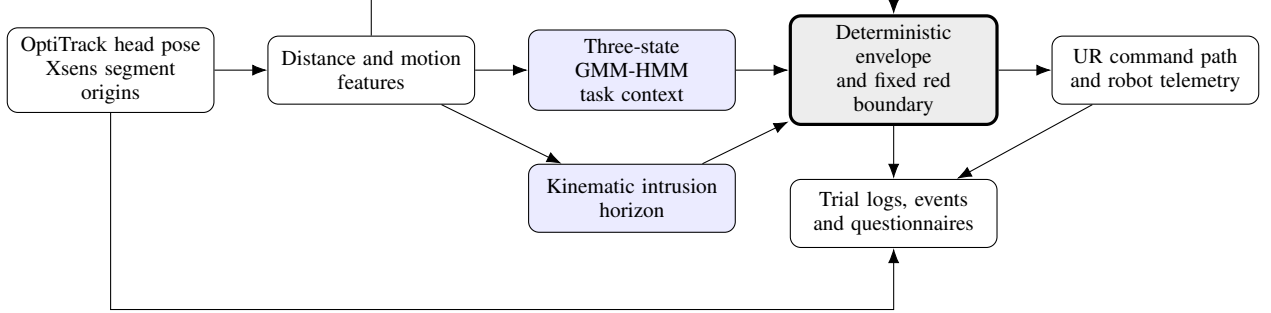


Fig. 1. Prototype architecture. Learned outputs (blue) can add caution but cannot raise the command above the deterministic envelope (grey). Sensing, decisions, commands and measured robot response are logged for every trial.



Fig. 2. Laboratory apparatus. (a) The cell: UR10 arm holding the panel surrogate at ceiling height, OptiTrack cameras on the overhead frame, and the operator desk. (b) A participant wearing the Xsens MVN suit and OptiTrack head cap, viewed from behind while approaching the robot. (c) The working phase: the participant works beneath the raised panel. Faces are obscured.

the closing speed in m/s; positive v_{proj} means the person is moving toward the robot. Heading alignment h_t , used later, is the alignment of the velocity vector with $\mathbf{u}$, not torso orientation. Table II collects the symbols used in this section.

Two geometry implementations were used during data collection. (Throughout this paper a code such as P17 denotes a study ID assigned to one recorded participant session, and T05 a trial within it; see Section V-B.) Sessions P17 and P22 use head features that are not arithmetically consistent with a same-row live-TCP column reconstruction. P24 records a live TCP reference for the head feature and an anchored set of segment origins for the controller’s body-distance input. The phase model uses head-derived features in both versions. The two versions are reported separately throughout.

D. Task Context and Intrusion Prediction

The three hidden phases are approaching, working and re-treating. The observation vector is $\mathbf{x}_t = [d_t, v_{\text{proj},t}, \|\mathbf{v}_t\|, h_t]^T$,

where h_t is velocity-heading alignment. Each state’s emission density is a three-component diagonal Gaussian mixture:

$$p(\mathbf{x}_t | z_t = i) = \sum_{k=1}^3 w_{ik} \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_{ik}, \text{diag}(\boldsymbol{\sigma}_{ik}^2)), \quad (2)$$

where z_t is the hidden phase at time t , w_{ik} is the weight of mixture component k for phase i (the three weights sum to one), and $\boldsymbol{\mu}_{ik}$ and $\boldsymbol{\sigma}_{ik}^2$ are that component’s mean and per-feature variance. The online forward filter uses present and past observations; offline Viterbi decoding also uses subsequent observations. Their validation values are therefore reported separately. Long labelled phases and the transition prior produce highly persistent states, which may delay recognition of short movements.

The intrusion channel operates independently of the HMM. It solves

$$d - r_{\text{red}} - v_{\text{proj}}\tau - \frac{1}{2}a_{\text{proj}}\tau^2 = 0 \quad (3)$$

TABLE II
NOTATION USED IN THE CONTROLLER EQUATIONS.

Symbol	Meaning
d	Separation between tracked point and robot proxy (m)
$v_{\text{proj}}, a_{\text{proj}}$	Closing speed (m/s) and closing acceleration (m/s ²)
h_t	Velocity-heading alignment with the robot direction
z_t	Hidden task phase at time t (approaching, working, retreating)
$w_{ik}, \mu_{ik}, \sigma_{ik}^2$	Weight, mean, variance of mixture component k , phase i
r_{red}	Radius of the fixed inner (red) boundary (m)
τ	Predicted time to reach the red boundary (s)
S	Protective separation distance (m)
T	Assumed reaction plus stopping time (s)
C	Intrusion allowance for body parts not tracked (m)
S_a	Allowance for sensing uncertainty (m)
$u_{\text{env}}, u_{\text{context}}, u_{\text{cmd}}$	Envelope, context, commanded speed fractions (0–1)

for the time τ at which the person, continuing at the current closing speed v_{proj} and closing acceleration a_{proj} , would reach the fixed inner boundary of radius r_{red} from the current separation d . A non-negative τ within a 0.5-s horizon triggers the intrusion state. The configured closing-speed gate is 0.6 m/s and projected acceleration is limited to ± 4 m/s². The predictor detects approach to the boundary; it does not infer intent.

E. Controller Policies and Command Path

The fixed controller maps constant distance bands to full speed, reduced speed and a zero request. The reactive controller uses closing speed in a simplified envelope. The predictive controller adds phase caution and intrusion prediction to that envelope. Its separation parameterisation is

$$S = \max(0, v_{\text{proj}})T + C + S_a, \quad (4)$$

where S is the protective separation distance the controller tries to keep, T is the assumed reaction plus stopping time so that $v_{\text{proj}}T$ is the distance the person closes before the robot stops, C is a constant intrusion allowance for parts of the body that the tracked point does not cover, and S_a is an allowance for sensing uncertainty. This is the form of the ISO/TS 15066 separation formula with the robot's own travel term omitted [2]. The configured values are $T = 0.4$ s, $C = 0.2$ m and $S_a = 0.1$ m. The ramp above S is 0.3 m. The fixed inner radius is 0.94 m; the fixed yellow radius is 1.6 times that value. These values were selected for the laboratory cell and are not measured stopping or uncertainty bounds of an installed system.

In SSM mode, the software constraint is

$$u_{\text{cmd}} = \min(u_{\text{env}}, u_{\text{context}}), \quad 0 \leq u_{\text{cmd}} \leq 1, \quad (5)$$

where u_{env} is the speed fraction allowed by the reactive envelope, u_{context} is the fraction proposed from phase context and intrusion prediction, and u_{cmd} is the fraction sent to the robot. The minimum means the learned context can only lower the command. A fixed inner-boundary override forces a

TABLE III
LEAVE-ONE-OPERATOR-OUT PHASE-RECOGNITION VALIDATION ON THE DEVELOPMENT SET. VITERBI DECODING USES THE FULL SEQUENCE; THE CAUSAL FILTER USES PRESENT AND PAST OBSERVATIONS.

Measure	Value
Development participants	3
Accepted trials	32
Labelled frames	90487
Offline Viterbi accuracy	0.821
Offline balanced accuracy	0.757
Causal-filter accuracy	0.794
Causal-filter balanced accuracy	0.725
Approaching recall	0.693
Working recall	0.923
Retreating recall	0.654
Offline fold P03 accuracy	0.812
Offline fold P04 accuracy	0.817
Offline fold P05 accuracy	0.832

zero request inside r_{red} , and unavailable tracking is handled conservatively by falling back to the most restrictive state. The command is applied through the UR speed override and the controller's pause interface. The prototype is a research controller: its command path is not a safety-rated protective stop, and the bound in (5) constrains the software request rather than certifying the installed system (Section VII-E).

IV. MODEL DEVELOPMENT AND VALIDATION

The phase classifier was developed before the participant study using 32 accepted labelled trials recorded by the three team members acting as operators under study IDs P03 (eight trials), P04 (12) and P05 (12), containing 90,487 frames. Class counts are 17,267 approaching, 53,609 working and 19,611 retreating. Validation used leave-one-operator-out cross-validation.

Two measures are reported. Accuracy is the fraction of all frames whose predicted phase matches the label; because working frames dominate, a model that favours the working class scores well on it. Balanced accuracy is the mean of the three per-class recalls, so each phase counts equally regardless of how many frames it has. Offline accuracy is 0.821 and causal accuracy is 0.794 (Table III). Balanced accuracy is lower, particularly for the causal filter (0.725), and the working class has the highest offline recall. The small development pool, temporal correlation within trials and the use of the same folds during development limit the generalisation of these values.

V. EXPERIMENTAL METHOD

A. Task and Experimental Design

The protocol specifies three blocks labelled A, B and C, with fixed, reactive and predictive conditions assigned through six rotated orders. Each block schedules one clean, one distractor and one rapid-movement trial. The ceiling-panel sequence includes approach and loading, retreat before lifting, work near the raised panel, and retreat before lowering and unloading. The scripted cue is placed during the lift and was delivered by the operator, so cue windows mark the instruction rather than the measured onset of movement.

B. Participants and Analysed Records

Twelve participants completed the protocol under Monash University ethics approval (project 51642). One further session was aborted for technical reasons and is excluded. Each session was assigned a study ID (P-number); because some sessions were restarted with a new ID, the IDs label recorded sessions and do not map one-to-one onto individuals.

A descriptive analysis was used because complete within-person controller sequences and consistent participant linkage were unavailable. Telemetry entered the analysis only where a session's sample stream and event journal were both readable and internally consistent (Section V-C); at the time of analysis this criterion was met by 14 trials from three sessions, P17, P22 and P24, and the remaining sessions contribute questionnaire data only. The analysis therefore included 14 telemetry trials associated with three study IDs and 31 block questionnaire submissions associated with 11 IDs. Nine IDs had all three block questionnaires and two lacked the final block response. Matched intake and end responses were available for all 11 follow-up IDs. Telemetry and questionnaire outcomes were analysed separately because their coverage differs. Recorded samples and original form responses were used directly; no missing values were imputed.

C. Telemetry Processing

A trial entered the telemetry analysis when its sample stream and event journal were readable and its sample count matched the trial metadata. Fourteen trials from the three sessions named above met these criteria. One additional trial record lacked its sample stream and was excluded. The resulting dataset contains 65,697 samples.

For timestamps t_i , time summaries use left-held values only when $0 < t_{i+1} - t_i \leq 0.25$ s. Longer intervals are reported separately and excluded from both numerator and denominator. The final sample has no following exposure interval. Fresh motion telemetry requires an available packet, six finite joint velocities and a source age between zero and 0.25 s. A maximum absolute joint velocity exceeding 0.005 rad/s is a diagnostic motion flag; counts at 0.001 and 0.01 rad/s are retained as threshold sensitivity checks.

Head-feature consistency is assessed by recomputing point-to-column distance from the same row's head position and live TCP; this is an internal consistency check rather than an external calibration. Cue-window extrema are the minimum distance and maximum closing speed within a window and need not occur at the same instant. Zero-command time includes intentional holds and is therefore reported as such rather than as false interruption.

D. Questionnaire Inclusion and Scoring

Questionnaire analysis excluded pilot, link-test and qualification submissions. There were 18 intake records, 31 block responses and 11 end responses after those exclusions. Block counts were A: 11, B: 11 and C: 9, with no duplicate ID/block keys.

The block questionnaire uses five-point ratings. Three semantic differentials run from anxious to relaxed, agitated to

calm, and quiescent to surprised. Four agreement items concern stopping confidence, constant monitoring, unnecessary stopping and speed legibility. Three adapted items concern mental demand, physical demand and frustration. The workload items are adapted from NASA-TLX [16] and the confidence and monitoring items from Jian et al. [17]; neither subset constitutes the full validated instrument, so items are analysed individually in their original direction and no composite score is formed.

Block results are presented as raw response counts and median [first, third quartile], using linear quantile interpolation. A sensitivity summary uses only the nine complete A–C response sets. Within these sets, counts of increases, ties and decreases between A and C retain the pairing and original item direction; they describe sequence-related changes without interpreting higher as uniformly better. Results are grouped by recorded block label rather than controller because assignment records alone do not establish the controller each questionnaire refers to. Intake/end comparisons match study IDs and retain the change from expected to experienced safety and trust. Both forms use 1 = not at all and 5 = completely.

Because neither the independent sample size nor complete controller exposure can be established, the analysis is descriptive: no significance tests, confidence intervals or effect estimates are reported, and missing observations are not assumed to be random.

VI. RESULTS

A. Telemetry Characteristics

The telemetry dataset contains five trials under P17, eight under P22 and one under P24. P17 and P22 also have block and end questionnaires; P24 has an intake record. Table IV summarises the recorded controller, scheduled event, capture span and zero-command proportion for each analysed trial.

The 14 analysed trials contain 65,697 samples and 23.13 minutes of summed capture spans (Table IV). 57 intervals exceed 250 ms, totalling 39.20 s or 2.82% of those spans; 1348.69 s remain for time integrals. In total 40 rows carry a stale-tracking flag. Controller decisions exist in 65,684 rows, of which 110 indicate unavailable tracking and invoke the conservative controller state.

Zone entries and stop onsets (Table IV) follow the task structure: every trial contains two to seven entries into the red zone, which correspond to the loading, working and unloading approaches the task requires, so red-zone entries are not a count of hazardous events. Stop onsets separate the controllers more clearly. Fixed and reactive trials contain six to nine protective-stop onsets each, of which three to five last at least 1 s. The four predictive trials contain 15, 20, 29 and 73 onsets, but only two or three per trial last at least 1 s; the remainder are stop segments shorter than 0.5 s in which the command alternated rapidly between stop and run. The predictive policy therefore did not hold the robot stopped for longer, but it switched the stop request far more often, and P24 T01, the only trial with the anchored segment-origin geometry, shows the most switching (68 segments under 0.5 s).

Across ten non-clean cue windows, fresh joint telemetry exceeds the diagnostic motion threshold for 236.31 s of

TABLE IV

ANALYSED TELEMETRY TRIALS. SPAN IS CAPTURE DURATION, NOT ISOLATED TASK TIME. ZERO-REQUEST TIME INCLUDES INTENTIONAL HOLDS. ZONE ENTRIES COUNT TRANSITIONS OF THE CONTROLLER ZONE STATE INTO YELLOW AND INTO RED; STOP ONSETS COUNT TRANSITIONS OF THE COMMAND INTO PROTECTIVE STOP, WITH THE NUMBER LASTING AT LEAST 1 S IN PARENTHESES.

Code/trial	Controller	Planned event	Rows	Span (s)	Gaps (s)	Zero request (%)	Yellow entries	Red entries	Stop onsets (≥ 1 s)
P17 T05	Fixed	Distractor	5,186	103.2	2.45	30.1	8	5	6 (5)
P17 T06	Fixed	Clean	4,550	91.7	2.55	30.8	8	5	9 (3)
P17 T07	Reactive	Clean	5,061	101.1	2.88	41.3	7	5	7 (3)
P17 T08	Reactive	Distractor	4,815	97.0	3.06	34.2	8	6	7 (5)
P17 T09	Reactive	Rapid Cue	4,426	92.1	2.50	32.8	9	5	9 (3)
P22 T02	Reactive	Distractor	5,415	108.3	2.31	34.4	10	5	8 (4)
P22 T03	Reactive	Rapid Cue	4,668	93.6	2.28	29.0	8	4	8 (4)
P22 T05	Predictive	Rapid Cue	5,315	109.0	3.05	34.9	8	4	20 (3)
P22 T06	Predictive	Clean	4,400	90.0	2.62	27.0	4	2	15 (2)
P22 T07	Predictive	Distractor	4,950	101.1	2.75	25.9	7	3	29 (2)
P22 T08	Fixed	Rapid Cue	4,745	96.0	3.13	31.4	9	5	8 (5)
P22 T09	Fixed	Distractor	4,772	95.6	2.58	24.9	7	3	8 (3)
P22 T10	Fixed	Clean	4,761	95.1	2.59	25.3	4	2	6 (2)
P24 T01	Predictive	Rapid Cue	2,633	114.2	4.45	39.9	13	7	73 (2)

Intervals over 250 ms are excluded from time integrals. Zero-request percentage uses the remaining positive intervals as its denominator. P17 and P22 use the earlier head geometry; P24 uses anchored segment origins for control.

observed intervals. The measurements confirm robot motion during labelled event windows. Among 13,048 fresh telemetry rows in these windows, motion counts are 11,509, 11,355 and 11,319 at 0.001, 0.005 and 0.01 rad/s, respectively. The presence of motion is therefore not sensitive to the chosen threshold.

B. Geometric Inputs and Cued Movements

The P17/P22 trial-level 95th-percentile discrepancies between recorded head distance and same-row live-TCP recomputation range from 0.413 to 0.534 m (Fig. 3a). P24's corresponding discrepancy is below 10^{-12} m, showing arithmetic agreement for its head reference. Arithmetic agreement establishes internal consistency of the P24 geometry; its physical uncertainty was not measured.

Recorded cue labels also differ from their intended kinematics. P17 T09 is labelled rapid intrusion, yet its minimum controller proxy is 2.034 m and maximum closing input is 0.388 m/s, below the configured gate. Conversely, the P17 T08 distractor window reaches a 0.496-m proxy and a 1.494-m/s maximum closing input. P22 T05 reaches a 0.252-m proxy during a rapid cue. These values are controller inputs rather than validated clearances, and they show that the cue label alone does not determine the kinematics that were actually recorded.

Figure 4 presents the recorded time series for P24 T01. Its rapid-cue window reaches a 0.913-m controller proxy and a 5.302-m/s maximum closing input. Command and joint-speed traces are presented together; because the cue marks the instruction rather than the movement onset, the delay between them is not reported as a protective response time.

C. Reported Experience

Block responses lean toward relaxed/calm, confidence in stopping, and low demand and frustration (Table V). Median stopping confidence is five in each block, while median unnecessary-stopping agreement is three in A and two in B/C.

The latter is a perception item rather than an objective false-stop count, and responses are clustered within study IDs.

Restricting the analysis to nine complete response sets preserves 29 of the 30 block-item medians. The exception is constant monitoring in B, whose median changes from two to one. The paired A–C counts nevertheless reveal variation hidden by marginal medians (Table VI). Five relaxed ratings increase and four remain unchanged; none decrease. Unnecessary-stopping ratings increase twice, decrease twice and remain unchanged five times, despite the lower C median. Surprise decreases for three codes, increases for two and is unchanged for four. Thus lower block medians do not establish a common within-code improvement.

Across 11 matched records, mean safety ratings were 3.82/5 before and 4.45/5 after the session. Eight ratings increased, one was unchanged and two decreased. Mean trust ratings were 4.00 before and 4.18 after; five increased, three were unchanged and three decreased (Table VII). The expected-to-experienced wording and incomplete individual linkage limit interpretation of these differences.

In end responses, C is selected as safest seven times, compared with two each for A and B. Full-shift choices are C: six, B: three and A: two. Least-safe choices are distributed across all labels (A: three; B: four; C: four). Since label-to-controller exposure is not fully established, C is not interpreted as the predictive controller. Four end responses report noticing changes between blocks and seven do not; the four comments refer to stopping or slowing sensitivity and to ascent speed. Median reported suit restriction is two (interquartile range 1.5–3), and no risk or discomfort comments were entered in the 11 end forms.

VII. DISCUSSION

A. Context-Aware Control in the Panel Task

The prototype integrates task context and kinematic prediction while preserving a deterministic software ceiling on the commanded speed. This architecture separates two questions: whether a movement belongs to a normal task phase, and

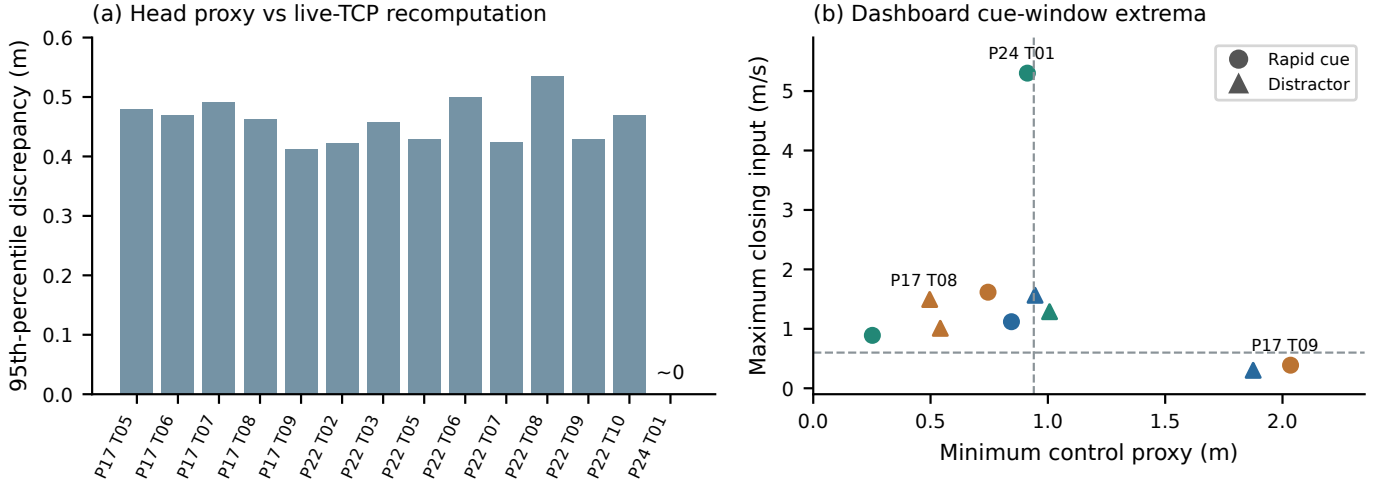


Fig. 3. Measurement diagnostics, one mark per trial. (a) For each trial, the 95th-percentile gap between the head distance the controller recorded and the distance recomputed from the same row’s head position and live TCP; P17 and P22 show gaps of 0.4–0.55 m, P24 shows none. (b) For each trial with a cue, the smallest control-proxy distance reached (horizontal) against the largest closing speed recorded (vertical) inside the cue window. Dashed lines show the configured 0.94-m inner boundary and 0.6-m/s closing gate: a rapid cue should land left of the vertical line and above the horizontal one. Colours identify fixed (blue), reactive (orange) and predictive (green) records. Geometry versions differ and the two extrema need not occur at the same instant.

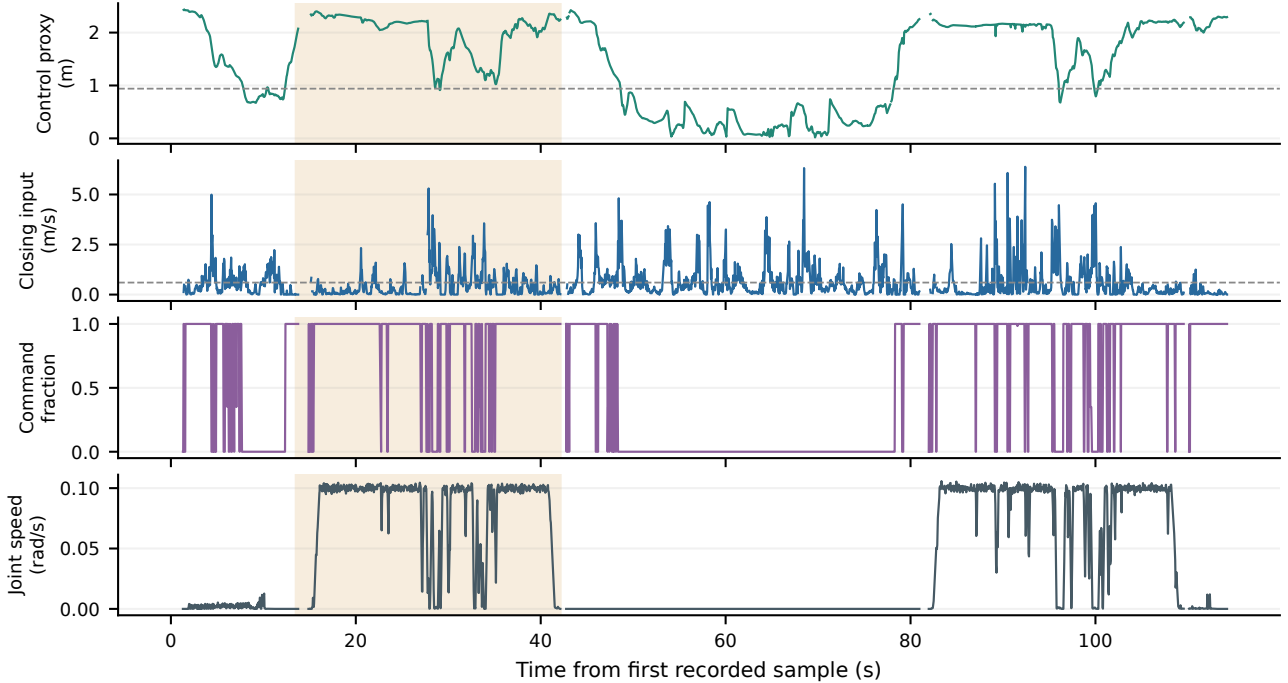


Fig. 4. P24 T01: recorded body-proxy distance, closing input, command fraction and maximum absolute fresh joint velocity. Shading marks the cue event window. Dashed lines are configuration thresholds. Gaps over 250 ms break the lines.

whether the measured trajectory approaches the configured boundary. An HMM misclassification need not authorise a higher speed than the geometric envelope, although errors in geometry or timing still affect that envelope.

The development results indicate that phase recognition is strongest during the longer working phase. The lower causal balanced accuracy compared with overall accuracy shows why phase accuracy alone is an insufficient measure of practical benefit. The stop-onset counts make the cost concrete: the predictive policy produced two to nine times as many

protective-stop onsets as the fixed and reactive policies in the same task, almost all of them sub-0.5-s segments in which phase caution and the intrusion predictor toggled the request near a threshold. This chattering is a controller-design defect rather than a property of the task, and it is largest under the segment-origin geometry, whose body-distance input is noisier than the head point. Hysteresis or a minimum stop dwell on the learned inputs would remove it without touching the deterministic envelope. The reactive reference is therefore an important comparator when assessing the added value of

TABLE V
BLOCK QUESTIONNAIRE ITEMS: MEDIAN [FIRST, THIRD QUARTILE]. A, B AND C ARE REPORTED BLOCK LABELS, NOT POOLED CONTROLLER CONDITIONS. ALL RATINGS USE 1–5.

Item	A (n = 11)	B (n = 11)	C (n = 9)
Anxious to relaxed	4 [3.5, 4.5]	5 [4.0, 5.0]	5 [4.0, 5.0]
Agitated to calm	4 [4.0, 5.0]	5 [4.0, 5.0]	5 [4.0, 5.0]
Quiescent to surprised	3 [1.0, 3.0]	3 [1.0, 4.0]	1 [1.0, 3.0]
Confident robot would stop	5 [4.0, 5.0]	5 [4.0, 5.0]	5 [4.0, 5.0]
Had to watch constantly	2 [1.0, 3.0]	2 [1.0, 3.0]	1 [1.0, 3.0]
Stopped more than needed	3 [2.0, 4.0]	2 [1.0, 3.5]	2 [2.0, 3.0]
Speed changes made sense	5 [3.5, 5.0]	5 [4.0, 5.0]	4 [4.0, 5.0]
Mental demand	1 [1.0, 3.0]	1 [1.0, 2.0]	1 [1.0, 2.0]
Physical demand	1 [1.0, 3.0]	1 [1.0, 2.5]	1 [1.0, 3.0]
Frustration	1 [1.0, 3.0]	1 [1.0, 2.0]	1 [1.0, 1.0]

Higher means the right-hand semantic anchor for the first three items, stronger agreement for the next four, and greater demand/frustration for the final three. Responses are clustered within 11 study IDs; individual-level linkage is incomplete. No trust, safety or workload composite is used.

TABLE VI
COMPLETE RESPONSE SETS: BLOCK MEDIANS AND WITHIN-CODE CHANGES (n = 9 CODES).

Item	A/B/C	A to C: up/same/down
Anxious to relaxed	4/5/5	5/4/0
Agitated to calm	4/5/5	3/6/0
Quiescent to surprised	3/3/1	2/4/3
Confident robot would stop	5/5/5	3/4/2
Had to watch constantly	2/1/1	1/5/3
Stopped more than needed	3/2/2	2/5/2
Speed changes made sense	5/5/4	1/5/3
Mental demand	1/1/1	2/6/1
Physical demand	1/1/1	1/6/2
Frustration	1/1/1	1/5/3

Counts refer to the raw rating direction, not uniformly to improvement. A–C changes describe block sequence, not a controller effect. The same nine codes supply every row.

TABLE VII
PAIRED GLOBAL RATINGS FOR 11 CODES WITH INTAKE AND END RESPONSES. FIVE-POINT ITEMS; DESCRIPTIVE CODE-LEVEL CHANGES ONLY.

Item	Before	After	Δ	Up/same/down
Felt safe	3.82	4.45	0.64	8/1/2
Trust in distance	4.00	4.18	0.18	5/3/3

Before/after wording changes from expected to experienced safety/trust. Means summarize coded responses; neither causation nor an interval-scale measurement model is assumed.

learned context.

Equation 5 also constrains the expected benefit. For identical SSM inputs, parameters and zone state, the predictive policy cannot request a higher speed than the reactive policy: prediction and phase caution can only preserve or reduce the request. Its incremental value would therefore need to arise from the timing of intervention or changes in subsequent human–robot coordination, rather than a more permissive instantaneous speed bound. Such a closed-loop benefit cannot be inferred from phase accuracy or from applying alternative policies to the same recorded trajectory.

B. Objective Behaviour and Perceived Experience

The telemetry illustrates changes in the distance input, speed request and measured joint motion throughout the panel se-

quence. The difference between geometric reference configurations materially affects how these signals can be interpreted. Similarly, variation between scheduled rapid movements and recorded closing inputs shows that event labels cannot substitute for measured exposure. These findings are consistent with the need to validate the sensing-to-stopping chain when implementing SSM [3].

Questionnaire responses generally indicate confidence in the robot stopping, low mental and physical demand, and limited frustration. Mean experienced safety exceeds mean expected safety in the matched records, whereas the trust difference is smaller. These observations suggest a positive reported experience of the laboratory task. However, novelty, supervised operation, repeated exposure and expectations could all contribute to the pattern. The change cannot be attributed specifically to the predictive controller.

The complete-response analysis makes the broad descriptive pattern less dependent on the two missing final-block forms, but does not remove bias from missingness or resolve individual linkage. Ceiling ratings for stopping confidence and floor ratings for demand/frustration also leave limited room to distinguish conditions. The paired stopping and surprise responses show why aggregate medians should be read alongside within-code changes. Increased familiarity is a plausible explanation for greater relaxation across blocks, but was not isolated by this analysis.

The end-of-session preference for block C does not provide a controller ranking: block labels rotate across controller assignments, and complete exposure linkage is unavailable. Likewise, subjective agreement that the robot stopped more than needed is informative about perceived interruption but is not an objective false-stop rate. Subjective and physical measures address different aspects of collaboration and should be interpreted together.

C. Safety and Efficiency Interpretation

Dynamic SSM research motivates evaluation of availability and cycle performance [8], [9]. The present study demonstrates the implementation needed to investigate those outcomes but does not establish a safety or efficiency advantage. Zero-command time combines intentional holds and responses to proximity; capture span includes operator actions and cannot

be equated with isolated task time. These quantities therefore do not establish false interruption, task efficiency or collaboration fluency [15].

A non-inferiority conclusion would additionally require a justified margin, validated protective geometry and end-to-end response measurements, none of which the current analysis provides; absence of a demonstrated difference is not evidence of equivalent safety.

D. Coverage of the Planned Measures

The study design specified four system measures (human–robot distance, number of speed reductions and stops, number of yellow/red zone entries and task completion time) and four experience measures (safety, comfort, predictability and trust) for comparison between static and adaptive safety. Distance, zone entries and stop onsets are reported per trial (Table IV, Figs. 3 and 4), with the caveat that red-zone entries and a share of the stop time arise from the loading and unloading approaches the task requires, so they are not counts of unnecessary interruption. Capture span is reported in place of task completion time because operator actions fall within the span. Safety and trust are reported as intake/end ratings, comfort through the relaxed and calm items, and predictability through the speed-legibility item. The static-versus-adaptive comparison remains open for the linkage and exposure reasons set out below.

E. Limitations

The sample is small and drawn by convenience. Session restarts prevented reliable linkage of every study ID to a distinct individual, so record counts cannot be treated as independent participant counts. Two questionnaire sets lack block C, and objective and subjective records cover different subsets of the study. Missingness may relate to technical problems or task execution, limiting complete-case interpretation and generalisation.

The geometric configuration changed during collection. Head points and segment origins do not represent complete body, tool and payload surfaces, and their agreement with a geometric calculation does not establish physical calibration. The controller is a research prototype: the separation envelope is simplified, the command path uses the ordinary robot speed override rather than a safety-rated stop, and no end-to-end protective response was measured, so the results do not support a certified safety claim for the installed system. Cued movements, operator pacing, visible motion, learning and fatigue may influence both behaviour and subjective reports. The model-development results are based on three operators and do not establish generalisation to construction workers.

F. Implications for Further Development

Future controlled evaluation should maintain a consistent geometry and validate its uncertainty against the complete moving volume. Synchronized measurements of physical movement, controller output and robot deceleration would allow response-time estimation independently of operator cues.

Task-phase estimates could then be evaluated for their incremental benefit over the reactive policy using verified within-person controller exposure. A persistent study identifier across restarted sessions would support that comparison without changing the participant-facing task.

VIII. CONCLUSION

This paper presented a context-aware speed and separation monitoring controller for human–robot ceiling-panel installation and evaluated it on a physical prototype with human participants. The controller combines measured closing kinematics, three-state task-phase recognition and kinematic intrusion prediction beneath a deterministic command ceiling, so that learned context can only make the robot more cautious. Causal phase recognition reached 0.794 accuracy in leave-one-operator-out validation. Across 14 telemetry trials and 31 block questionnaires, participants reported high confidence in the robot stopping and low workload, and perceived safety rose over the session; the telemetry showed that geometric reference selection and cue execution govern how recorded control behaviour should be interpreted. Complete-response analysis preserved 29 of 30 block medians, while paired changes showed that lower median interruption ratings were not uniform within participants. Establishing a controller-specific safety or efficiency advantage requires the consistent geometry, synchronised response measurement and persistent participant linkage identified here, and these define the next stage of the work.

DATA AVAILABILITY

Analysis scripts, derived tables and figure-generation code accompany the manuscript. Participant-level data require controlled access. All reported experimental summaries are calculated from recorded observations without imputation or substitution of simulated values.

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AI tools assisted with data-processing code, descriptive analysis, figure production and manuscript drafting and revision.

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